

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: CSA14

COURSE NAME: Compiler Design

COURSE OUTCOME COVERED

CO4: Examine intermediate code generation techniques to enhance code optimization (BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 4

Total Marks:40

Annexure A

CLASS TEST

Code of Conduct:

I Vishnu G / 192424086 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Vishnu G

Q. No	Understanding of Concepts (25)	Explanation (25)	Application (20)	Organization (15)	Accuracy(10)	Total Score (out of 100)
1	/ 4	/ 4	/ 4	/ 4	/ 4	
2	/ 4	/ 4	/ 4	/ 4	/ 4	
3	/ 4	/ 4	/ 4	/ 4	/ 4	
4	/ 4	/ 4	/ 4	/ 4	/ 4	
5	/ 4	/ 4	/ 4	/ 4	/ 4	
OVERALL RUBRICS VALUE						
	/ 4	/ 4	/ 4	/ 4	/ 4	
	/ 25	/ 30	/ 20	/ 15	/ 10	/ 100

Annexure A

CLASS TEST

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	15%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps

G. Vishnu

192424086

CSA 1407

CO4-AT1 Class Test

Compiler Design

1) if $((a < b) \text{ and } ((c < d) \text{ or } (a > d)))$ then

$z = x + y * z$

else

$z = z + 1$

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$((a < b) \text{ AND } ((c < d) \text{ OR } (a > d)))$

Write assembly code of the statement

L1: if $a < b$ goto L2
goto L5

L2: if $c < d$ goto L4
goto L3

L3: if $a > d$ goto L4
goto L5

L4:
 $t_1 = y * z$
 $t_2 = x + t_1$
 $z = t_2$
goto L6

L5:
 $t_3 = z + 1$
 $z = t_3$

L6:

2) Translate $(a < b)$ and $(c < d)$ or $(e < f)$ into three address statement using

if $a < b$ goto 3.

Backpatch statement 2 to statement 5.

Backpatch statement 4 to statement 5.

if $a < b$ goto 3

goto 5

if $c < d$ goto 7

goto 5.

if $e < f$ goto 7

goto 8

TRUE

FALSE

Backpatching means generating jump instruction first and filling their target address.

3) $A = (-A + B) * ((-A + B) * c)$

$$t_1 = -A$$

$$t_2 = t_1 + B$$

$$t_3 = -A$$

$$t_4 = t_3 + B$$

$$t_5 = t_4 * c$$

$$t_6 = t_2 * t_5$$

$$A = t_6.$$

	Operator	arg 1	- arg 2	Result
T ₁	unminus	A	-	t ₁
T ₂	+	t ₁	B	t ₂
T ₃	unminus	A	-	t ₃
T ₄	+	t ₃	B	t ₄
T ₅	*	t ₄	C	t ₅
T ₆	*	t ₂	t ₅	t ₆
T ₇	=	t ₆	-	A.

4) $B \rightarrow B_1 \parallel B_2$
 $B \rightarrow B_1 \& \& B_2$
 $B_1 \rightarrow !B_1$
 $B \rightarrow E_1 \text{ relop } E_2$
 $B \rightarrow \text{true}$
 $B \rightarrow \text{false}$

L₁ :
 if $A \neq B$ goto LTRUE
 goto L~~TRUE~~₂

L₂ :
 if $C < d$ goto LTRUE
 goto L₃

L₃ :
 if $e < f$ goto LTRUE
 goto LFALSE.

LTRUE:

goto LFEND.

LFALSE:

Boolean expression is false.

LFEND: